Progress Report

- Increment 2 -

Group Hoo's Fishin'

1) **Team Members**

   Juan Pablo Pastrana II - JP25 - JuanPablo-29

   Tyler Dussault - tjd22 - TylerDussault

   Bridget Gizzi - bag23c - gizzibridget-ops

   Darius Davis - dew20 - Hevi-D

   Dhruv Patel - drp24 - DhruvP16

2) **Project Title and Description**

   Hoo's Fishin'

   Hoo's Fishin' is a 2D fishing game where the player controls an owl named Hoo and catches fish using a fishing mechanic inspired by Stardew Valley. The player can sell the fish they catch to a shopkeeper and use the money to buy better equipment. As the player continues fishing and earning money, they can improve their gear, catch rarer fish, and unlock more areas of the map.

3) **Accomplishments and overall project status during this increment**

   During this increment, we focused on expanding the gameplay systems developed during Increment 1 and implementing more of the features from our original proposal. We added a backpack system that allows players to purchase additional storage space and manage items between their backpack and hotbar. We also implemented player tasks and catch tracking, giving players fishing objectives to complete for rewards while keeping track of the fish they have caught. Along with these additions, we improved the fishing rod upgrade system, added a main menu and day/night cycle, and expanded the fishing mechanics with a new fish minigame. These features helped build on the basic fishing and selling gameplay loop by adding more ways for players to progress through the game.

   Another accomplishment was improving the functionality and stability of the existing systems. We made changes to the fishing minigame so that rod upgrades have a greater effect on gameplay, and we improved the shop and hotbar interfaces. We also fixed several bugs involving player interactions, inventory management, and overlapping interface elements. These improvements were important because the new features needed to work with the systems already implemented during Increment 1.

   Compared to our original proposal, the project has made good progress toward the planned functionality. The core gameplay loop of catching fish, storing them, selling them, and purchasing upgrades was completed during Increment 1, and this increment expanded on that foundation by adding more inventory management, player objectives, and equipment progression. Several features that were planned for Increment 2, including the main menu, backpack, improved fishing mechanics, and player tasks, have now been implemented. However, some parts of the original scope, such as bait, lures, achievements, and additional areas of the map, are still incomplete. Overall, the project is moving closer to the original vision, with most of the main gameplay systems in place and the remaining work focused on expanding content, completing progression features, and improving the game before the final increment.

4) **Challenges, changes in the plan and scope of the project and things that went wrong during this increment**

During this increment, one of our main challenges was identifying and resolving pre-existing bugs in the codebase. We addressed this by carefully investigating each issue to find its root cause and fix it. Although the process was tedious, it was essential to making sure the project stayed stable and worked as intended.

Adding to this challenge, much of our work involved building on code written by other teammates, which we didn't always fully understand. Adding new features and improving parts of the game often meant changing code we weren't familiar with, and that made both development and debugging harder. Fortunately, our team communicates well. Through regular meetings and frequent check-ins, we were able to ask questions, clarify how different parts of the code worked, and keep making progress.

5) Team Member Contribution for this increment

Juan Pablo Pastrana II:

a) I wrote the stakeholder email in the progress report.
b) I did not contribute to the requirements and design document in this iteration.
c) I did not contribute to the implementation and testing document in this iteration.
d) I fixed bugs that had to do with restricting the player from interacting with multiple systems at once, for example only being able to fish and not open the shop or backpack. Also, only being able to use the shop or backpack individually. Fixed shop UI overlays. Refactored inventory code so that when new fishes are added they do not need a hardcoded if/else statement by using a dictionary to have a reference to the item sprite. Finally, I added a flappy bird fish with a new minigame.
e) I did not contribute to the video in this iteration.

Tyler Dussault:

a) My team carried me through the progress report. I was not left with anything to write.
b) As with the requirements and design document, all the work was done by the time I was ready to contribute.
c) I did not contribute to the testing document.
d) Most of my contributions for this increment were visual improvements. I added a dynamic day/night cycle, fireflies that flicker at night, reflection and shimmering of the water, and a boat for the owl to sail in. I also added fishing sound effects to make the world feel more alive.
e) For the video, I provided a demo of the current build of our game, showcasing all the new added features.

Bridget Gizzi:

a) I contributed to section 4, the challenges and changes to plans and scope section.
b) I did not contribute to the requirements and design doc in iteration 2.
c) I did not contribute to the requirements and design doc in iteration 2.
d) In this iteration, I created the main menu for the game right when its opened. This included directly matching the background of the game with its multiple layers of images. Additionally making sure stylistic choices worked well with the rest of the game like creating a semi transparent background and functional buttons to be able to start the game and exit. This menu can be accessed at any point in the game if the user needs to quit. I additionally spent time debugging issues that arose in the creation of this such as placement of the buttons and menu on the screen, in comparison to the rest of the game screen.
e) I did not contribute to the video in this iteration.

Darius Davis:

a) I contributed information about the fishing rod improvements completed in increment 2, including how higher-level rods now make the fishing minigame easier for players.
b) I contributed to the implementation of the fishing rod requirement that higher-level rods should provide more room for error during the fishing minigame. The updated system uses different catch-zone multipliers depending on the required fishing rod.
c) I performed functional testing on all three fishing rod levels in Godot. I verified that the Basic Rod maintained its original catch-zone size, the Improved Rod provided a larger catch zone, and the Advanced Rod provided the largest catch zone. I also tested that the catch zone still moved correctly, the player could successfully catch fish, and the minigame continued working without script errors.
d) For the source code, I worked on improving the fishing rod system and how it affects the fishing minigame. I added a new catch_zone_multiplier property to the FishingRod resource so different rods could provide different amounts of room for error. I configured the Basic Rod to use the normal catch-zone size, the Improved Rod to have a 15% larger catch zone, and the Advanced Rod to have a 30% larger catch zone. I also updated the fishing minigame script to adjust the catch-zone size based on the equipped rod. I connected this functionality through the fish spawn manager so that the minigame receives the correct multiplier when the player starts fishing. I made sure the visible catch zone matched the actual catching area and prevented the zone from becoming larger than the fishing track. These changes allow upgraded rods to make fishing easier while keeping the existing catch-speed bonuses.
e) I was responsible for presenting the challenges and issues encountered during Increment 2, along with our team's plans and goals for Increment 3. I discussed the difficulties of integrating new features, the improvements made to address gameplay issues, and the remaining features we plan to implement in the next increment.

Dhruv Patel:

a) Wrote the accomplishments and overall project status section, including the features implemented during Increment 2 and how the project compares to the original proposal.
b) Worked on the requirements and design document, updating the functional and non-functional requirements to reflect Increment 2 features. Documented the backpack, rod upgrades, catch tracking, and player tasks. Also updated the class and use case diagrams to reflect the new systems and their relationships.
c) Wrote the programming language, technologies used, functional testing, non-functional testing, and non-execution-based testing sections. Documented the use of GDScript and Godot, how gameplay features are tested by running the game, and how GitHub, group chat discussions, meetings, code reviews, and walkthroughs are used to review changes.
d) Worked on the backpack, fishing rod upgrades, catch tracking, and player task systems. Implemented the backpack system so players can purchase additional inventory space, store items, and swap items between the backpack and hotbar. Worked on simplifying the fishing rod upgrade system so that rods and their upgrades are managed in one place instead of requiring changes in multiple files when adding new rods or upgrades. Implemented a catch tracking system that keeps track of the total number of fish caught, the number caught for each fish species, and whether a species has been discovered. Also worked on the player task system, allowing players to accept fishing tasks, track their progress based on fish caught, and receive money when tasks are completed. Added a tasks tab to the existing shop interface so players can switch between shopping and viewing available tasks. Worked on adjusting the shop and tasks interface so both tabs have a consistent appearance and do not change size when switching between them.
e) I went over the general overview of the project and gave a description of what was accomplished during this increment and state of our project in the video.

6) Plans for the next increment

- Adding bait and lure items to provide more fishing options.
- Implementing an achievement system to track player accomplishments and reward progress.
- Adding more fish types, fishing rods, and equipment.
- Expanding the shop with additional items and upgrades.
- Adding new areas of the map for players to unlock and explore.
- Improving the fishing minigame and balancing fish difficulty and rod upgrades.

7) Stakeholder Communication

Hoo's Fishin' Progress Report Two

Hello, this is a follow up on what has been accomplished in the development of the game "Hoo's Fishin'" since last time.

There have been many new improvements in this sprint. Some of the highlights are a task tab in the shop that allows the player to get additional money for completing specific tasks, a new backpack upgrade that lets the player hold more than two species of fish, and also new vibrant visuals that help build the aesthetic especially with the day and night cycle. One new fish has been added called the Flappy Fish that has a different fishing minigame. Some polish that was introduced is the new main menu screen to greet players as soon as they open the game and the rods make a difference when upgraded to how difficult the fishing minigame is. Along with this there were many under the hood optimizations to allow for more dynamic integrations of systems and expansion.

Some setbacks we experienced were balancing some of the new mechanics with each other, and figuring out solutions to get requirements fit into the game loop. Art is also a difficult area so lots of prototyping is just colored boxes for now.

What we are looking forward to getting to is the bait and lure system, and ways to promote the player to explore more of the map. More fish are coming with creative takes and a final boss fight to add an ultimate goal the player can achieve.

Best regards,

The Hoo's Fishin' Team

8) Link to video

Increment2Video.mp4